Name: Savitha R

Reg.no: 192421368

6 Write a program to implement Naïve Bayes algorithm in python and to display the results using

confusion matrix and accuracy.

Program:

```
# Naive Bayes Classification
# Confusion Matrix and Accuracy

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

# Dataset
# [Age, Salary]

X = [
   [22, 20000],
   [25, 25000],
   [28, 30000],
   [30, 35000],
   [35, 40000],
   [40, 45000],
   [45, 50000],
   [50, 55000],
   [23, 22000],
   [27, 28000],
   [32, 36000],
   [38, 42000],
```

```
    [43, 48000],
    [48, 52000]
]

# 0 = Not Purchased
# 1 = Purchased

y = [
    0,
    0,
    0,
    0,
    1,
    1,
    1,
    1,
    0,
    0,
    1,
    1,
    1,
    1
]

# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
```

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)

# Create Naive Bayes model
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Display actual and predicted values
print("Actual Values    :", y_test)
print("Predicted Values :", y_pred)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
```

output:

```
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Actual Values    : [0, 1, 0, 1, 1]
Predicted Values : [0 1 0 1 1]

Confusion Matrix:
[[2 0]
 [0 3]]

Accuracy: 1.0
Accuracy Percentage: 100.0 %
```